
From Detection to Decision: A Lightweight Framework for Autonomous Driving

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Abstract

This paper explores a lightweight, rule-based GO/STOP decision framework for urban autonomous driving, based on a RetinaNet object detector and simple heuristic logic. We apply the system to the KITTI dataset to evaluate how visual object detection outputs can be used for straightforward decision-making in structured environments. The objective is not to pursue state-of-the-art performance, but rather to build an interpretable and modular pipeline for prototyping and learning purposes. The results suggest that this approach can serve as a starting point for integrating perception and control in early-stage autonomous driving systems.

1 Introduction

Autonomous driving requires not only perception of the environment but also decisions about motion. This work investigates a lightweight decision-making system using RetinaNet for object detection and a rule-based engine to determine whether to proceed or stop. The goal is to demonstrate a simple visual decision pipeline that may support experimentation or educational use, without claiming technical novelty or competitive performance.

2 System Overview

The system combines two stages: (1) object detection using RetinaNet and (2) decision logic based on manually defined rules. Detected objects are analyzed to determine if they represent an obstacle that requires stopping. The modular design makes the system easy to inspect, adjust, and build upon.

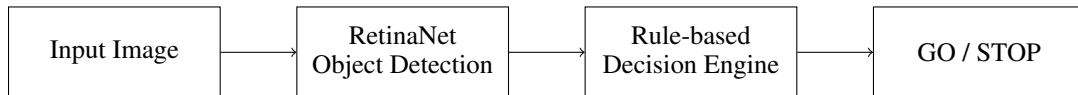


Figure 1: System-level overview: From image to GO/STOP decision.

3 Related Work

Object detection models such as YOLO, SSD, Faster R-CNN, and RetinaNet are widely used in autonomous systems. RetinaNet (1) uses a focal loss to address class imbalance in dense detection tasks. Rule-based systems, while limited in flexibility, are common in prototyping scenarios due to their transparency and simplicity.

4 Dataset and Preprocessing

We use the KITTI dataset (2), which includes annotated driving scenes from real-world urban environments. The images are resized and normalized before being input to the RetinaNet model. Object annotations are also used to automatically derive GO/STOP labels for evaluation, based on the same rules as the inference stage.

5 RetinaNet-based Detection Pipeline

RetinaNet uses ResNet50 as the backbone, extracting feature maps at multiple levels. These maps feed into a Feature Pyramid Network (FPN), which generates multi-scale features used by detection heads.

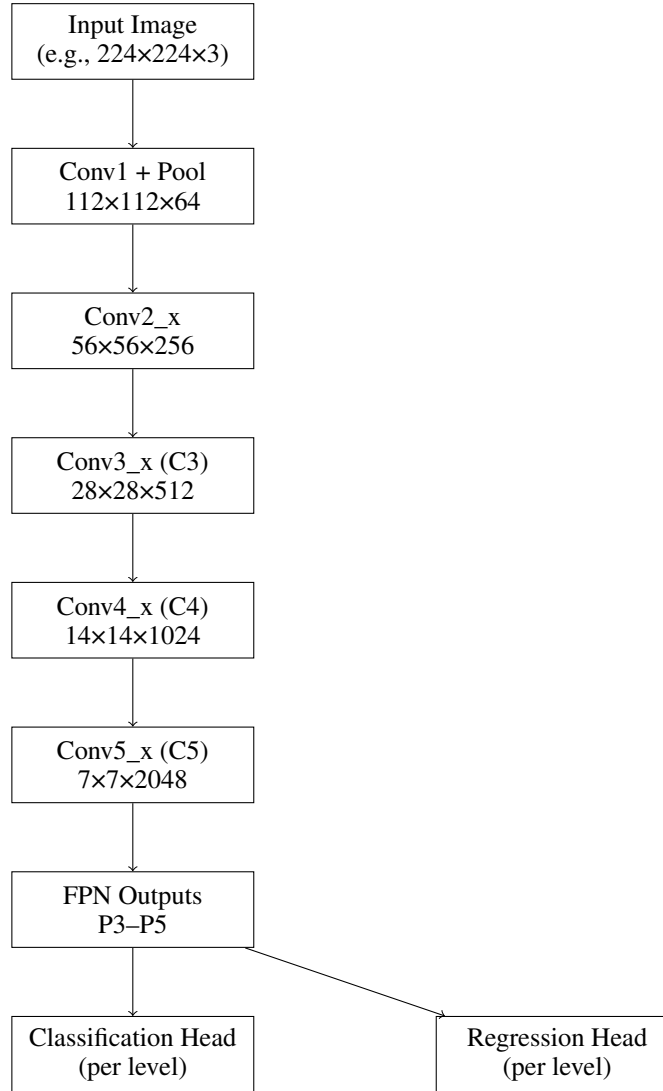


Figure 2: RetinaNet architecture with feature map sizes. FPN outputs P3–P5 are processed by classification and box regression heads.

This hierarchical architecture enables robust object detection at multiple scales, suitable for identifying both large vehicles and smaller pedestrians in traffic scenes.

6 Decision Logic

We define a rule-based engine that outputs GO or STOP based on the detections:

- If a pedestrian is detected, STOP.
- If a vehicle (car, truck, van, or bus) exceeds a certain bounding box size (e.g., 300px), STOP.
- Otherwise, GO.

```
for each detection:
    if class == "pedestrian":
        return STOP
    if class in ["car", "truck", "van", "bus"] and (w >= limit or h >= limit):
        return STOP
return GO
```

7 Evaluation

We generated labels by applying the same rule-based logic to KITTI annotations. Evaluation was carried out on:

- A 10-image test set: 90% accuracy.
- A 100-image random sample from KITTI: 85% accuracy.

This suggests that simple decision logic, when paired with a reasonably accurate detector, can produce acceptable results in well-structured settings.

8 Limitations

While the proposed framework offers interpretability and simplicity, it has several limitations:

- **No temporal reasoning:** The decision logic operates on single frames without considering motion continuity, leading to instability in dynamic scenarios.
- **Static threshold rules:** The use of fixed bounding box size thresholds may not generalize well to varying camera angles, object distances, or lighting conditions.
- **Binary output only:** The system only supports GO/STOP decisions. It does not account for nuanced behaviors like slowing down, lane changing, or yielding.
- **No contextual semantics:** The system does not understand lane structure, traffic light states, or road signs. It only reacts to bounding box size and class label.
- **Detector dependency:** Any misclassification or localization error by RetinaNet directly propagates to the decision output without correction or verification.

9 Future Work

Several directions can extend the capabilities of this framework:

- **Temporal integration:** Incorporating frame sequences, object tracking, or recurrent models (e.g., ConvLSTM) may improve decision stability under motion.
- **Learning-based decision modules:** Instead of static rules, a lightweight classifier or reinforcement learning agent could learn decision boundaries from labeled data or simulation environments.
- **Multi-action decision space:** Extending the system to support more actions such as turning, slowing down, or signaling would better reflect real-world driving logic.
- **Semantic context integration:** Combining object detection with semantic segmentation (e.g., road layout, lanes) or visual cue recognition (e.g., traffic lights, signs) can enhance decision precision.

- **Error-aware decision logic:** Introducing uncertainty modeling (e.g., via confidence thresholds, MC dropout) or rule fallback mechanisms can make the system more robust to detector failures.
- **Real-time deployment on embedded devices:** Optimizing the framework (e.g., model quantization, pruning) for edge deployment would help evaluate its feasibility in practical systems.

10 Conclusion

We implemented a simple GO/STOP decision framework using object detection and rule-based logic. The design emphasizes clarity, modularity, and ease of experimentation, rather than performance. While limited in flexibility, such a system can provide a foundation for further development in interpretable decision-making pipelines for autonomous vehicles.

References

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